Summer Internship Summary

Title: Smart Irrigation and Monitoring System using ESP32 and RTL/SoC Design

Organization: Innovation and Incubation Hub MNNIT Foundation (IIHMF), MNNIT Allahabad, Prayagraj.

I completed a four-week summer internship (15 June 2026 to 14 July 2026) at the Innovation and Incubation Hub MNNIT Foundation (IIHMF), MNNIT Allahabad, under the guidance of Prof. Basant Kumar, Professor, Department of ECE, MNNIT Allahabad.

Objective: Build an automated, sensor-driven irrigation controller on the ESP32 with cloud monitoring, and re-implement its control algorithm as verified digital hardware (Verilog RTL) taken through synthesis and an RTL-to-GDSII flow.

Work Highlights:

- Built an ESP32 prototype with soil-moisture, DHT11, rain and ultrasonic tank-level sensors and a relay-driven pump, with automatic and manual modes and Blynk web and mobile dashboards.
- Designed the same algorithm as modular Verilog RTL (environment monitor, decision logic, timer and FSM), verified it with a self-checking testbench (8 of 8 tests passed, full FSM coverage) and synthesized it in Vivado (25 LUTs, 12 flip-flops).
- Extended the design through an RTL-to-GDSII flow with OpenLane 2 on SKY130, closing timing at 5 ns with zero DRC, LVS and antenna violations, and confirmed the netlist by gate-level simulation.

Key Learnings: Embedded firmware and IoT cloud integration, RTL design and self-checking verification, and an ASIC implementation flow (synthesis, static timing analysis, DRC/LVS signoff) using open-source tools.

Outcome: The internship gave end-to-end experience from an IoT prototype to a verified and laid-out hardware design, and showed how one algorithm can be validated in software and then realized in hardware.

Mentor Details

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